

Michael Xu

<https://michaelxu.dev>

Email: mxu1925@gmail.com

Linkedin: <in/yunhuixu>

EXPERIENCE

Senior Research Engineer

Sept. 2024 – Present

Microsoft Research - Deep Learning Group

San Francisco, CA

- Researched and optimized a novel generative flow-based RL objective for LLM post-training that outperformed GRPO on mathematical reasoning and code generation benchmarks, training end-to-end with verifiable rewards using verLL.
- Scaled on-policy RL post-training to 100B+ parameter Mixture-of-Experts models across multi-node B200 GPU clusters, leveraging Megatron-LM with expert parallelism and enabling asynchronous training to pipeline rollout generation and reward computation stages.
- Built containerized code execution environments for RL training on code generation, enabling safe parallel rollout evaluation with automated test-suite feedback as verifiable reward signals.
- Designed and implemented a scalable human-in-the-loop data collection platform on Xbox Insider, processing 50,000+ daily preference annotations through gamified crowdsourcing with built-in quality control, generating training data for reward modeling and pluralized alignment.

Research Engineer

Feb. 2021 – Sept. 2024

Microsoft Research - Natural Language Processing

San Francisco, CA

- Provided technical leadership for LLM integration efforts across Xbox game studios, implementing LLM-driven systems that enhanced narrative complexity and created rich generative player interactions in multiple in-development game titles.
- Cross-functional collaboration with Microsoft AI on the Gaming Copilot Companion initiative, integrating on-device multimodal vision models with real-time gameplay analysis to provide contextual assistance across multiple game genres, resulting in improved player experience and accessibility.
- Released a dynamic branching narrative framework using LLMs, enabling procedurally generated dialogue trees that increased story variation while maintaining narrative coherence across storylines ([paper](#)).
- Developed a structured conversational prompting framework to improve LLM alignment that improved quest scenario completion rates by 25% ([paper](#)).
- Communicated gaming AI research to executive stakeholders, driving strategic decision-making and securing continued investment in AI initiatives at Microsoft Research and Xbox.

PROJECTS

Craft an Iron Sword | *Minecraft, Conversational Agent, Tool Use, RAG* | [GitHub](#)

Aug. 2021 – Present

- Pioneered an agentic AI system for Minecraft combining LLMs with composable tool calling and dynamic RAG retrieval, through Javascript execution and promise chaining; predating modern tool-use paradigms ([paper](#)).
- Extended the system with multimodal capabilities including real-time video and voice interaction; showcased by CEO Satya Nadella at Build 2024 ([video](#)) and winner of Executive Hack at Microsoft Hackathon 2022.

Cryptopoly | *Typescript, Vue, Three.JS, WebRTC, Solidity* | [GitHub](#)

Apr. 2019 – Jun. 2023

- Designed and implemented a blockchain-based gaming protocol using Ethereum and State Channels, reducing transaction costs by 90% while maintaining decentralized gameplay integrity.
- Developed a full-featured Monopoly implementation as proof-of-concept, demonstrating transparent game mechanics and trustless multiplayer interactions on the blockchain.
- Engineered a scalable serving architecture handling concurrent gameplay for thousands of users, with built-in DDoS protection and automatic load balancing.

EDUCATION

University of Illinois at Urbana-Champaign

Champaign, IL

M.S. in Computer Science, Data Science Concentration - 3.91 GPA

Arizona State University

Tempe, AZ

B.S. in Computer Science, Minor in Statistics - 4.0 GPA

TECHNICAL EXPERTISE

Research Areas: Reinforcement Learning & Post-Training, Reward Modeling & RLHF, LLM Alignment & Evaluation, Multimodal AI Systems, Conversational AI & NLP

Languages: Python, TypeScript / JavaScript, SQL, C++, Java

ML & Infrastructure: PyTorch, verLL, vLLM, Transformers, TensorRT, Kubernetes, Docker, Azure